

**Aliah University**  
**Odd-Semester (Autumn) Examination - 2023**  
**(For 5<sup>th</sup> Semester, BTech. CSE)**

Paper Name: Formal Languages and Automata Theory  
 Paper Code: CSEUGPC13

Full Marks: 80  
 Time: 3 Hrs

**Group – A**

**1. Short answers –**

(5X2=10)

- Why Type-1 or CSL grammar is called “context-sensitive”? Give an example.
- Define Language. How many languages are possible for any alphabet  $\Sigma = \{a, b\}$ ?
- Prove that – “Regular languages are closed under intersection operation”.
- Define Recursively Enumerable and Non-Recursively Enumerable languages.
- What is “left recursion” in grammar? How is it removed?

**Group – B**

(Answer any 6 questions)

(6X5=30)

- What do you mean by the ‘power’ of machines in theory of computation? Compare different machines as per their power. [1+4]
- Construct Deterministic Finite Automata for the language below- [5]  
 $L$ : The set of all strings, not starting with ‘ba’, where  $\Sigma = \{a, b\}$ .
- Use Pumping Lemma to prove – Language  $L$  (given below) is not Regular. [5]  
 $L$ : The set of all strings with equal number of ‘a’s and ‘b’s, where  $\Sigma = \{a, b\}$ .
- Convert the Mealy Machine (given below) into Moore Machine. [5]

|               | Input = 0  |        | Input = 1  |        |
|---------------|------------|--------|------------|--------|
| Present State | Next State | Output | Next State | Output |
| A             | A          | 0      | B          | 0      |
| B             | B          | 1      | C          | 0      |
| C             | B          | 0      | C          | 1      |

- Construct a Turing Machine for Language  $L$  (given below). [5]  
 $L$ :  $\{a^i b^j \mid i = j\}$ .
- Write a Context Free Grammar for Language  $L$  (given below). [5]  
 $L$ :  $\{a^i b^m c^j \mid i = m + n \text{ and } n, m, i > 0 \text{ where } \Sigma = \{a, b, c\}\}$ .
- In the following context-free grammar, the set of terminal symbols is  $\Sigma = \{a, b, c\}$ , the set of non-terminal symbols is  $N = \{S, U, V\}$ , and the start symbol is  $S$ . Convert the grammar to the Chomsky normal form. Show all the steps of your conversion. [5]

$S \rightarrow U \mid V$   
 $U \rightarrow c \mid aVa \mid \epsilon$   
 $V \rightarrow c \mid bUb \mid \epsilon$



Group – C  
(Answer any 4 questions)

(10X4=40)

9. Consider the language below to answer the following questions –

$L$ : The set of all strings containing either even number of 0's or odd number of 1's but not both,  $\Sigma = \{0, 1\}$ .

- a) Construct a minimized DFA for  $L$ .
- b) Using Arden's theorem, find out the regular expression for  $L$ . [5+5]

10. Consider the language below to answer the following questions –

$L = \{ww^R \mid 'w' \text{ means a string formed over } \Sigma = \{a, b\} \text{ and } 'w^R' \text{ indicates the reverse of string 'w'}\}$

- a) Construct a Pushdown Automata (PDA) that accepts  $L$ .
- b) Write a Context Free Grammar (CFG) which generates  $L$ . [6+4]

11. Construct an  $\square$ -NFA for  $(0 + 1)^*(00 + 11)(0 + 1)^*$  and Convert to DFA. [3+7]

12.

- a) Construct a Transition diagram for the Turing Machine to accept the following language.  
 $L = \{\sigma \tau \sigma \mid n \geq 1, \Sigma = \{0, 1\}\}$
- b) Is it a deterministic or non-deterministic Turing machine? [8+2]

13. Discuss closure properties of Context-Free Languages (with proof) under Union, Intersection, Complement, Concatenation and Kleene-star operations. [5\*2]

14.

- a) Show that the following CFG is ambiguous-  $S \rightarrow iCtS \mid iCtSeS \mid a$   
 $C \rightarrow b$
- b) Construct a Push Down Automata for Language  $L$  (given below). [5+5]  
 $L = \{a^n b^m c^{2m} d \mid n, m > 0 \text{ where } \Sigma = \{a, b, c, d\}\}$ .